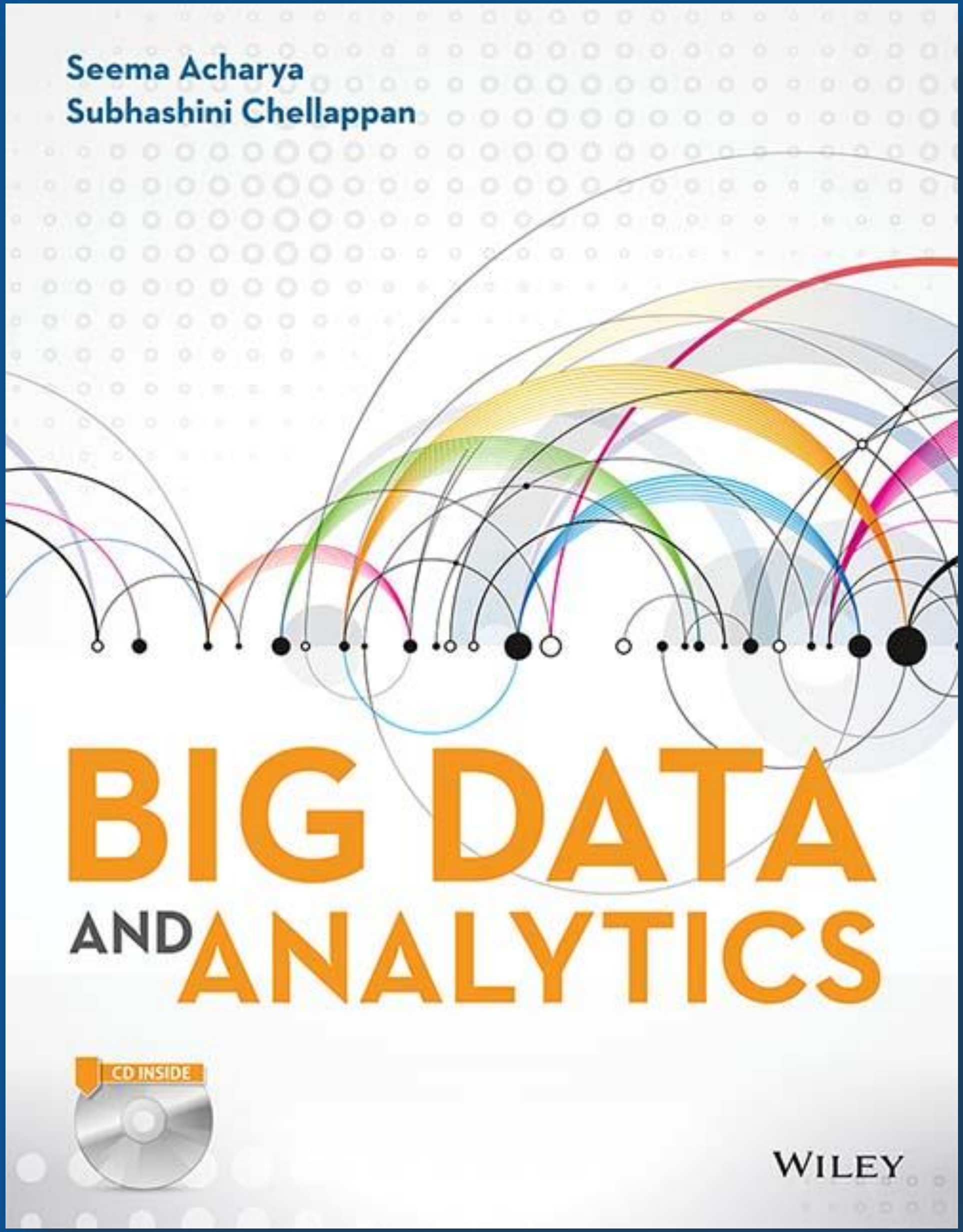


Big Data And Analytics

Seema Acharya
Subhashini Chellappan

Chapter 3

Big Data Analytics

Learning Objectives and Learning Outcomes

Learning Objectives	Learning Outcomes
Big Data Analytics	
1. What is big data analytics and what it isn’t?	a) To understand the significance of big data analytics.
2. Why is big data analytics important?	b) To understand the role of data scientist.
3. What is data Science?	c) To understand the various terminologies used in the big data environment.
4. Getting familiar with the terminologies used in the big data environment.	

Session Plan

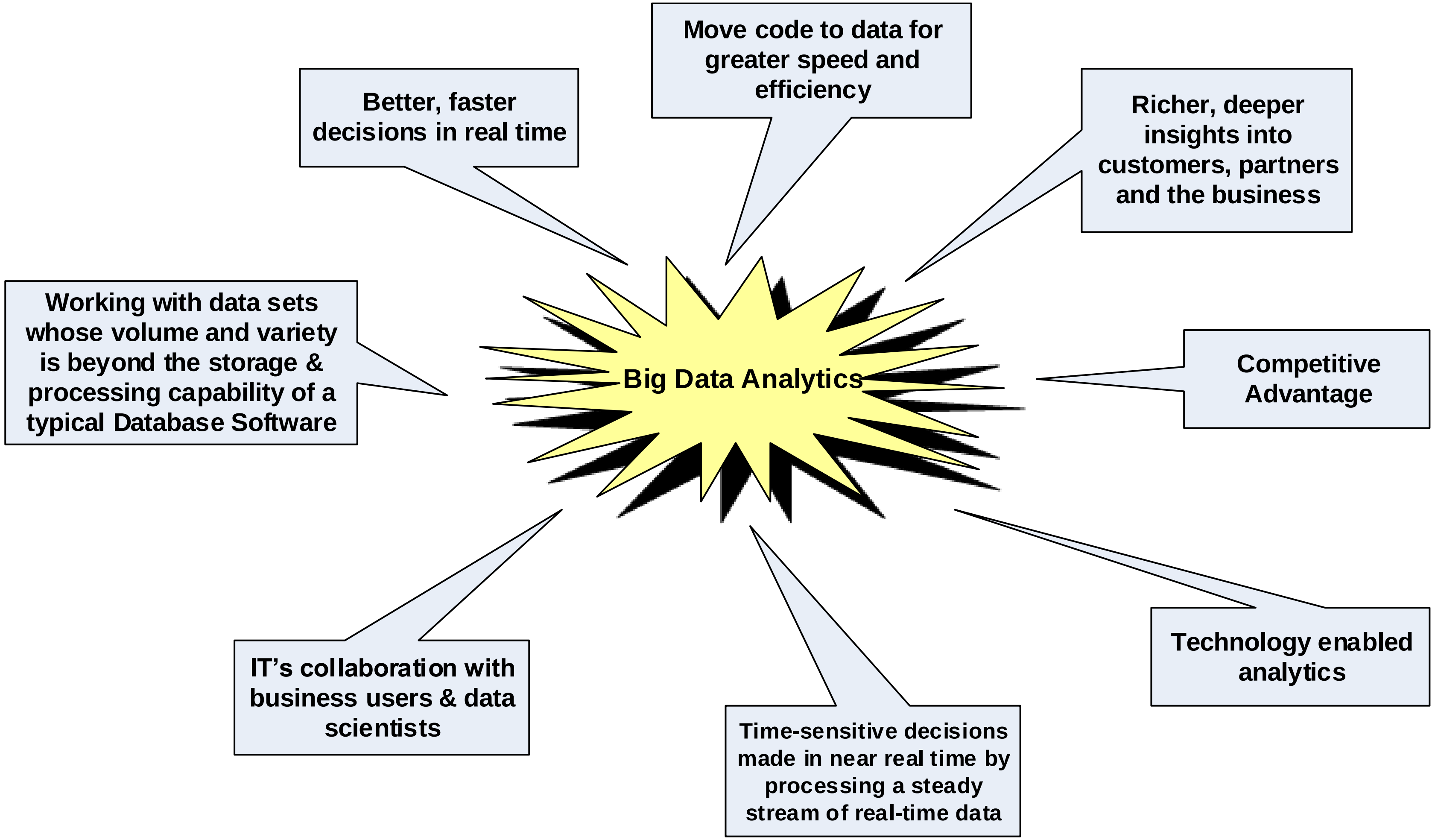
Lecture time	45 to 60 minutes
Q/A	15 minutes

Agenda

- What is Big Data Analytics?
- What Big Data Analytics isn’t?
- Classification of Analytics
- Why is Big Data Analytics Important?
- Data Science
- Data Scientist … Your New Best Friend!!!
- Terminologies Used in Big Data Environment
  - In Memory Analytics
  - In Database Processing
  - Massively Parallel Processing
  - Difference between Parallel versus Distributed Systems
  - Shared Nothing Architecture
  - Consistency, Availability, Partition Tolerance (CAP): Theorem Explained
- Few Top Analytics Tools

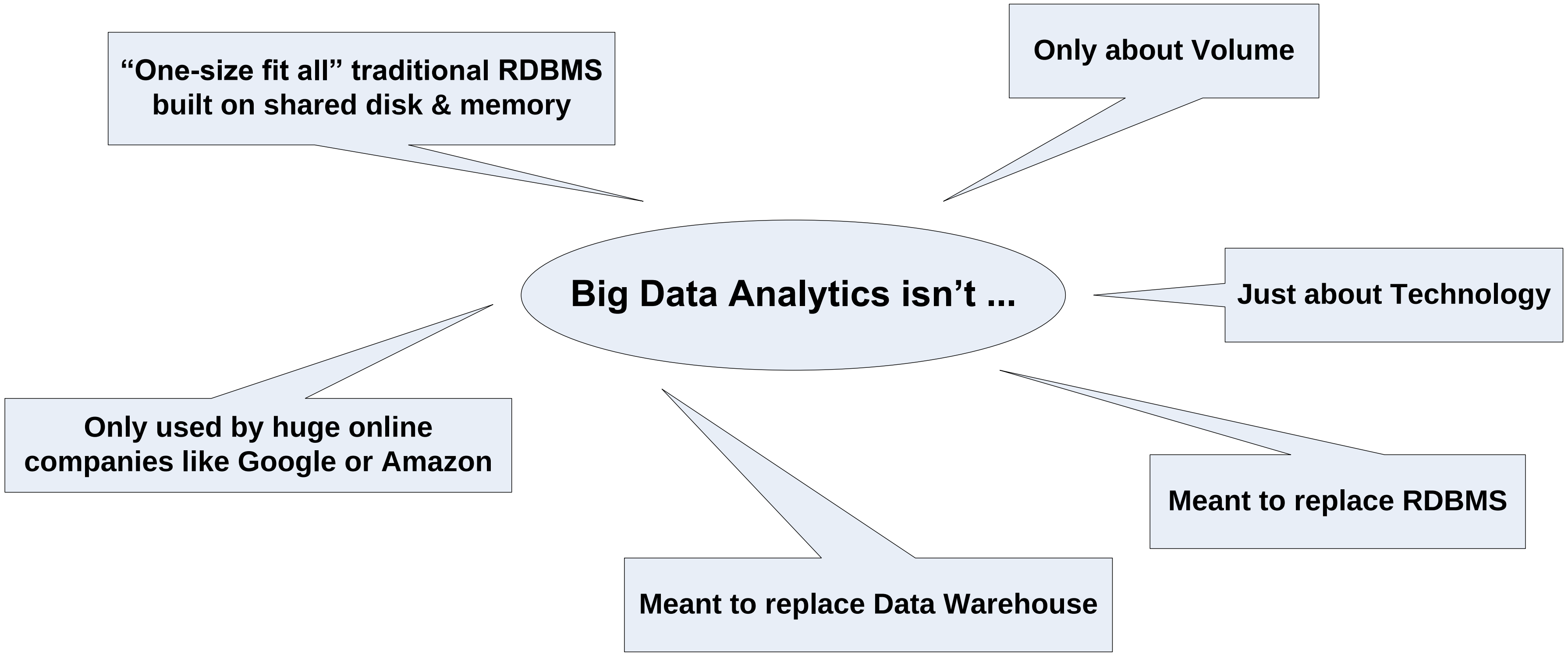
What is Big Data Analytics?

What is Big Data Analytics?



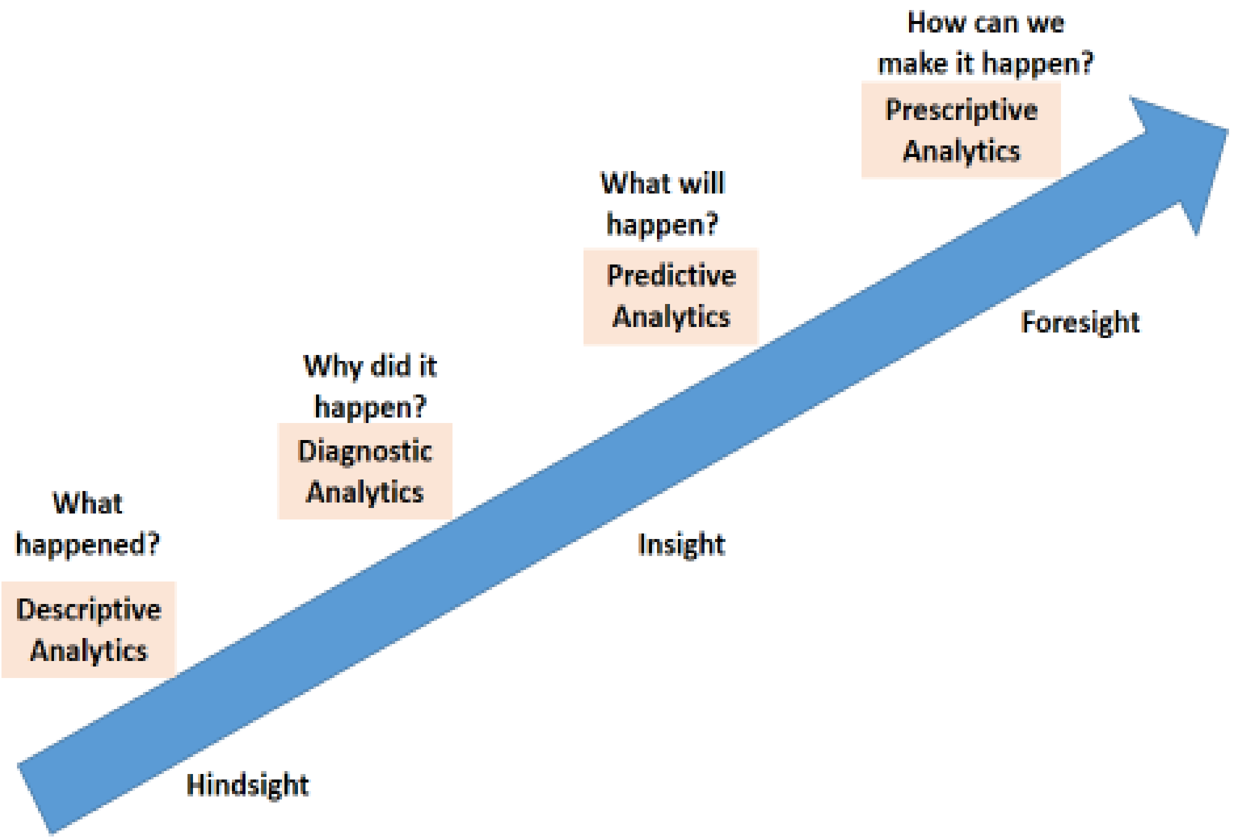
What Big Data Analytics isn't?

What Big Data Analytics isn't?



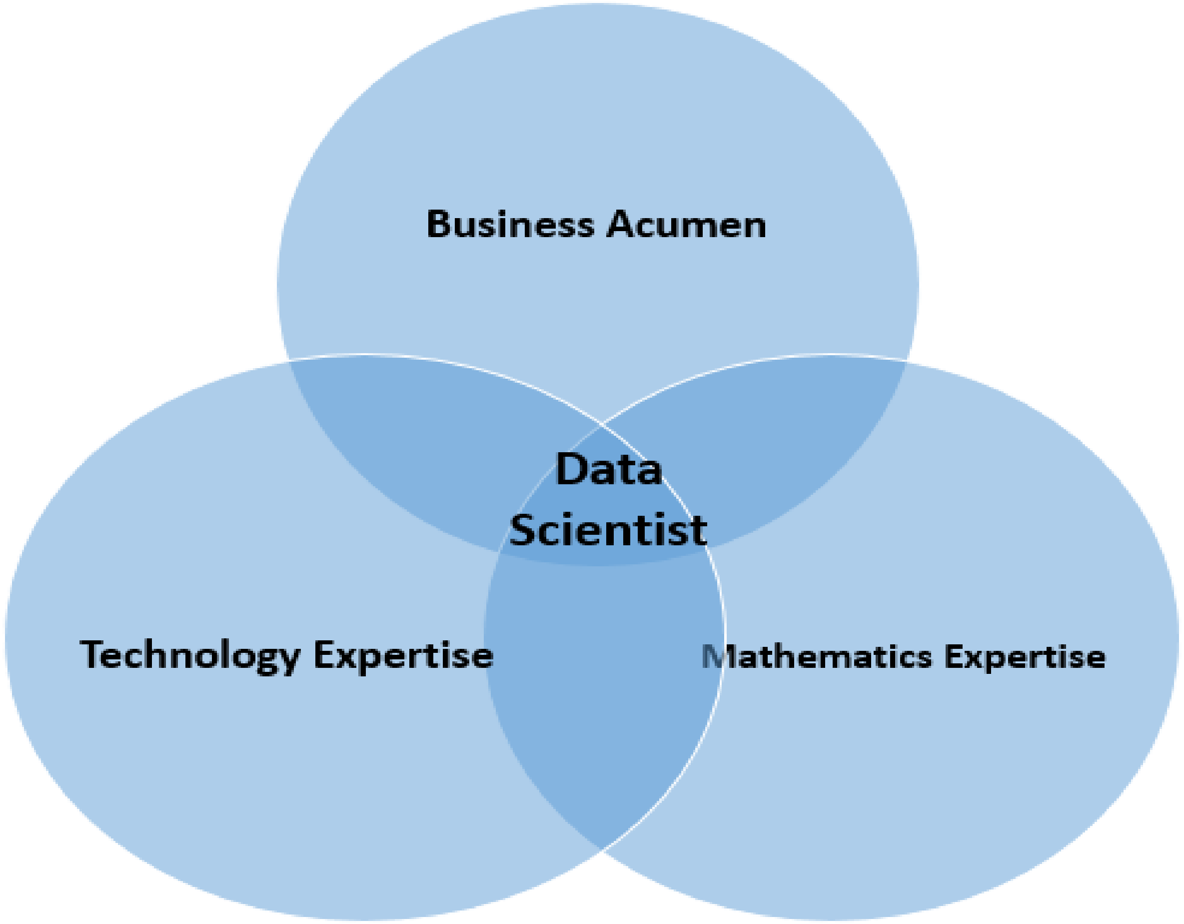
Analytics 1.0, 2.0 and 3.0

Analytics 1.0, 2.0 and 3.0



Data Science

Data Scientist

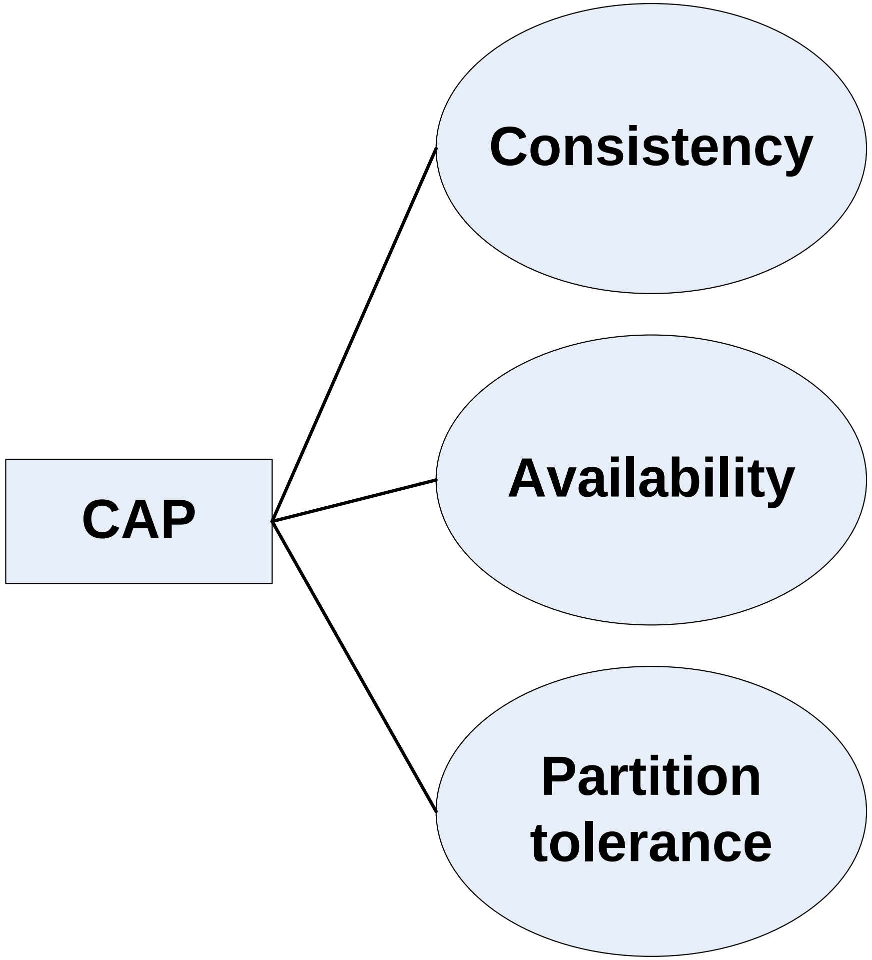


Terminologies Used in Big data Environments

- In-Memory Analytics
- In-Database Processing
- Massively Parallel Processing
- Parallel System
- Distributed System
- Shared Nothing Architecture

CAP Theorem

Brewer's CAP



Few Top Analytical Tools

Few Top Analytical Tools

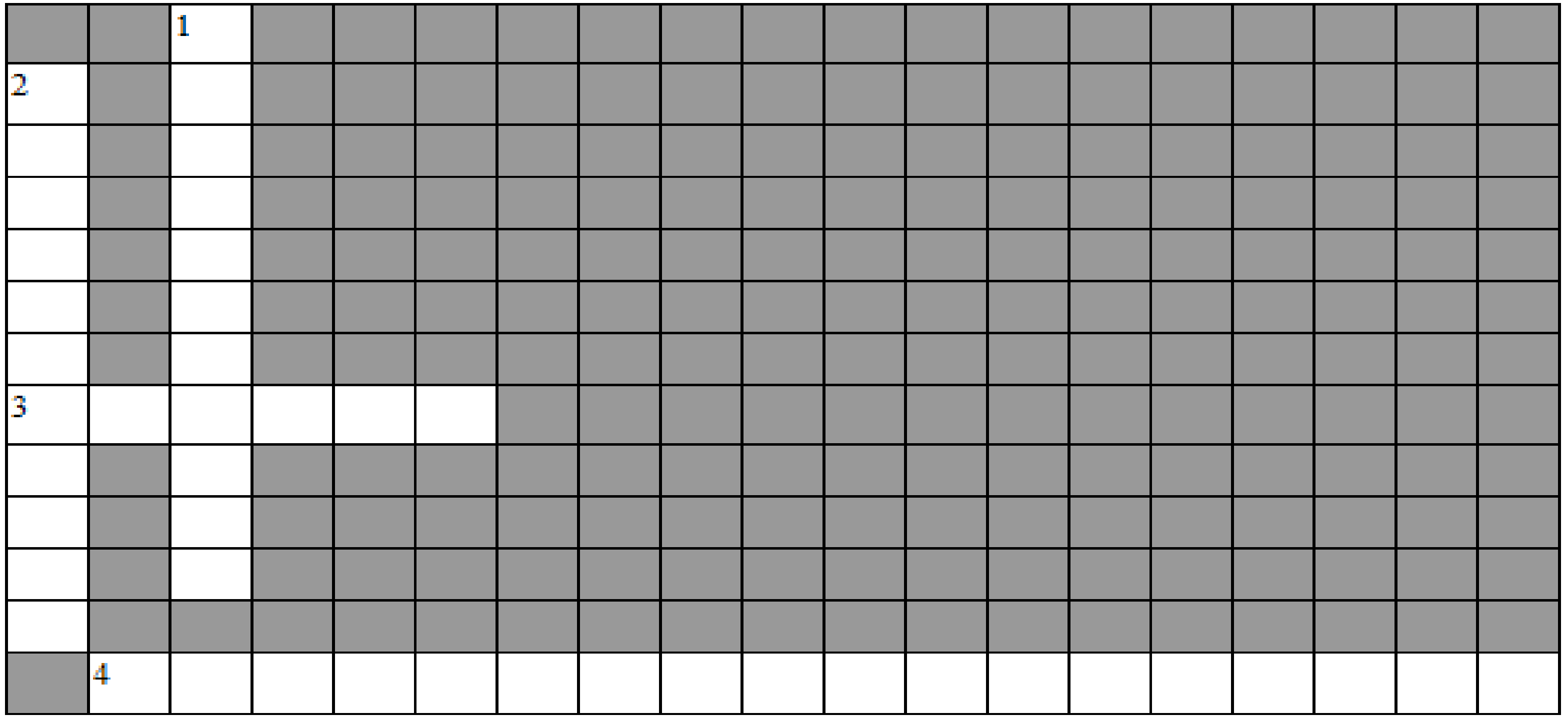
- MS Excel
  https://support.office.microsoft.com/en-in/article/Whats-new-in-Excel-2013-1cbc42cd-bfaf-43d7-9031-5688ef1392fd?CorrelationId=1a2171cc-191f-47de-8a55-08a5f2e9c739&ui=en-US&rs=en-IN&ad=IN

- SAS
  http://www.sas.com/en_us/home.html

- IBM SPSS Modeler
  http://www-01.ibm.com/software/analytics/spss/products/modeler/

Answer a few quick questions …

Crossword Puzzle on CAP theorem



ACROSS

3. CAP theorem is also called as ------------------ theorem.
4. System will continue to function even when network partition occurs.

DOWN

1. Every read fetches the most recent write
2. A non-failing node will return a reasonable response within a reasonable amount of time

Answer Me

- What are the key questions to be answered by all organizations stepping into analytics?
- What is predictive and prescriptive analytics?

Summary please…

Ask a few participants of the learning program to summarize the lecture.

References …

Further Readings

- http://en.wikipedia.org/wiki/Data_science
- http://simplystatistics.org/2013/12/12/the-key-word-in-data-science-is-not-data-it-is-science/
- http://www.oralytics.com/2012/06/data-science-is-multidisciplinary.html

Thank you